

## Artifacts and Evidence

## Artifacts and Evidence I

This section reports what the project has actually produced and how each quantitative claim is grounded. Every number below is substituted at render time from the deterministic producer, which reads the repository state at commit 57905ac5e, so each figure here is regenerated from the artifacts it describes rather than transcribed.

## Model-Family Coverage I

GNN ships a curated corpus of model families that exercise the language across the difficulty gradient from minimal parser fixtures to full multi-agent and scaling studies. The manifest registers 9 families (basics, discrete, continuous, hierarchical, multiagent, precision, structured, gridworld, scaling-study) across 9 target directories, and `input/gnn_files` holds 10 corpus directories containing 29 concrete example models. The three sets are not coextensive, and the manuscript does not treat them as one. Registered in no family: `input/gnn_files/learning` (1 of the 10 corpus directories), covered in sec. 7. Registered but outside the scanned tree: the multiagent family's target directory, whose model the gates exercise even though it is not among the 29 models under `input/gnn_files`. Each family declares the simulation

frameworks it is meant to drive, which is what lets the same text model fan out across the executable-model leg of the Triple Play (Smékal and Friedman 2023). The families and their declared frameworks are enumerated below.

# Model-Family Coverage I

Table 1: Model families declared in `input/model_family_manifest.json` and the frameworks each family targets. Capability splits in the Description column are generated from `src/render/framework_registry.py`, not authored in the manifest.

| Family        | Frameworks                         | Description                                                                                                                                                                                                                                                  |
|---------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| basics        | pymdp                              | Minimal perception fixtures used for parser and validator smoke coverage.                                                                                                                                                                                    |
| discrete      | pymdp                              | Discrete POMDP and HMM-style active inference fixtures.                                                                                                                                                                                                      |
| continuous    | jax, numpyro, stan, rxinfer        | Continuous-state linear-Gaussian fixtures (F/H/Q/R + Gaussian prior; continuous_navigation closes the loop on beliefs). Native on RxInfer.jl, JAX, PyTorch, NumPyro, Stan; reported unsupported (not failed) on PyMDP, ActiveInference.jl, DisCoPy, bnlearn. |
| hierarchical  | pymdp, rxinfer, jax                | Hierarchical and temporal model fixtures (per-level matrices composed into a joint POMDP for categorical backends; RxInfer.jl renders two-level models natively).                                                                                            |
| multiagent    | rxinfer                            | Multi-agent coordination and swarm fixtures.                                                                                                                                                                                                                 |
| precision     | pymdp                              | Precision weighting and curiosity-driven fixtures.                                                                                                                                                                                                           |
| structured    | pymdp                              | Structured factor graph and posterior fixtures.                                                                                                                                                                                                              |
| gridworld     | pymdp, rxinfer, activeinference_jl | Gridworld POMDP fixture used for cross-framework acceptance checks.                                                                                                                                                                                          |
| scaling-study | pymdp                              | PyMDP scaling-study fixtures, sampled conservatively for acceptance.                                                                                                                                                                                         |

## Model-Family Coverage I

The family-by-framework structure is shown in fig. 1, which renders the coverage matrix directly from the family registry rather than from a hand-maintained table.

# Model-Family Coverage I

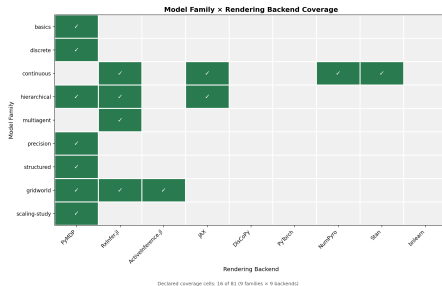


Figure 1: Model-family coverage across simulation frameworks, generated from the GNN family registry.

## Model-Family Coverage I

These families are not illustrative prose: they are the inputs over which the parser, the type checker, and the cross-framework code generators are exercised, and they are the substrate for the reliability gates described next.

## Semantic-Fidelity and Cross-Framework Reliability Gates I

Two reproducible-by-command gates check that GNN's promise — one text model, many faithful executable renderings — survives contact with real backends. Both live under `scripts/` and read the same family corpus described above, so they verify the artifacts the manuscript actually references.

The semantic-fidelity gate, `scripts/run_semantic_fidelity_gate.py`, checks that a model parsed from GNN text and then re-emitted preserves its semantic content: the state-space structure, the factor and modality declarations, and the matrix shapes implied by a discrete Active Inference generative model survive the round trip (Da Costa et al. 2020). It is meant to be run as a command and to report fidelity per model, not as a static claim baked into prose.

## Semantic-Fidelity and Cross-Framework Reliability Gates I

The cross-framework reliability gate, `scripts/run_cross_framework_reliability.py`, takes a single GNN model and renders it across multiple simulation backends, then checks that the resulting executable models agree on the structure they were generated from. The reference comparison runs on the continuous family across JAX, NumPyro, RxInfer.jl — 3 independent Active Inference engines spanning the Python and Julia ecosystems (Heins et al. 2022). The family declares 4 backends (JAX, NumPyro, Stan, RxInfer.jl); Stan is declared but not among the 7 frameworks the gate profiles, so it is excluded from the comparison. Because the same source model drives all 3 renderings, disagreement between backends localizes a generator defect rather than a modeling choice.

## Semantic-Fidelity and Cross-Framework Reliability Gates I

Both gates are stated here as commands you can run, not as asserted pass counts. The manuscript deliberately does not quote a fixed number of passing checks: the authoritative, current result is whatever those scripts report when executed against the corpus, and binding a frozen count into prose would invite exactly the drift the auto-injection contract exists to prevent.

## Repository Scale I

The repository's scale is itself evidence of the surface that the gates and pipeline cover, and it is reported in fig. 2 directly from the tracked files at commit 57905ac5e.

# Repository Scale I

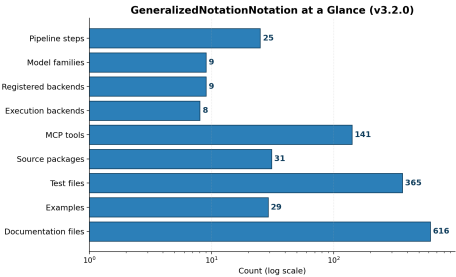


Figure 2: Repository-scale metrics — source packages, test files, and tool surface — measured from the tracked files at the commit the producer stamps.

The test suite comprises 365 test files containing 4103 test functions, exercising a source base of 582 Python files across 31 packages (194646 lines of source). The Model Context Protocol surface — which exposes GNN's capabilities to external agents and tools — provides 141 tools across 32 modules. The pipeline itself runs as 25 steps (0–24), and 105 figure artifacts from the rendering of figures, models, and reports are committed under `output/`, of which 6 are the manuscript's own.

A claim is manuscript-ready only when it is bound to a verifiable artifact. Concretely, every claim in this manuscript must rest on one of four support types:

## Claim Discipline I

- ▶ A passing test or validator command — for example the semantic-fidelity and cross-framework gates above, which can be re-run on demand.
- ▶ A generated output produced by a deterministic producer, such as the figures rendered from the family and repository scans, or the token values emitted by the manuscript-variable producer that backs every number on this page.
- ▶ A source ledger, manifest, or configuration file that fixes the value being claimed.
- ▶ A resolved entry in `references.bib` for any external-literature claim (Friston 2010; Parr et al. 2022).

## Claim Discipline I

The pipeline records its own evidence trail under `output/`. The run-level summary is written to `output/PIPELINE_REPORT.md`, and the per-step execution record — including which steps ran, their status, and their artifacts — is captured in `output/00_pipeline_summary`. These are the artifacts a reader should consult to confirm that the numbers substituted into this section correspond to a real, reproducible pipeline run rather than to asserted prose. When a value would otherwise need a literal number with no producer behind it, the discipline is to omit the number rather than to hard-code it.